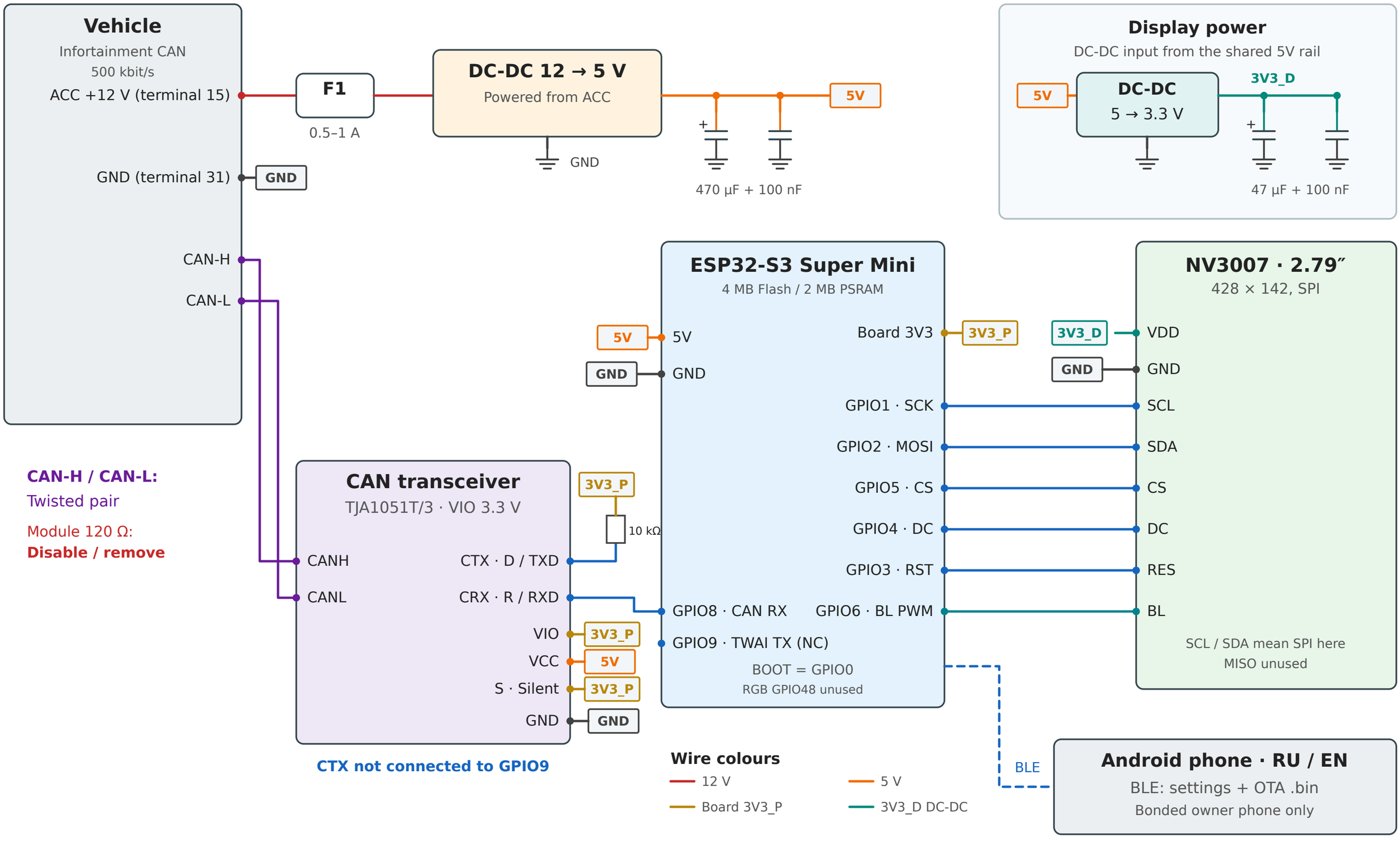
HUD — ESP32-S3 Super Mini + NV3007 2.79″

428 × 142 · Firmware GPIO mapping · Display via 5 → 3.3 V DC-DC · TJA1051T/3 CAN module

Based on gateway_supermini.svg · Receive-only CAN · Phone BLE settings and OTA



Connections and notes

All GNDs common. LCD VDD → 3V3_D from 5 → 3.3 V DC-DC; input → 5V. CAN power from ESP32-S3: VCC → 5V; VIO → board 3V3.
BL / GPIO6 is a logic-level backlight control. If BL directly powers LEDs, use an external driver.
CAN: CRX → GPIO8; GPIO9 reserved for TWAI, left unconnected. CTX pulled up to 3V3_P through 10 kΩ: receive only.
Capacitors: 100 µF + 100 nF at board 3V3; 100 nF at CAN VCC; 47 µF + 100 nF at LCD VDD.
Use TJA1051T/3 with VIO: S → VIO (Silent). A version without /3 and with 5 V logic cannot be connected directly.
BOOT (GPIO0): hold for 5 seconds, then release. Pins are shown logically, not in their physical board positions.